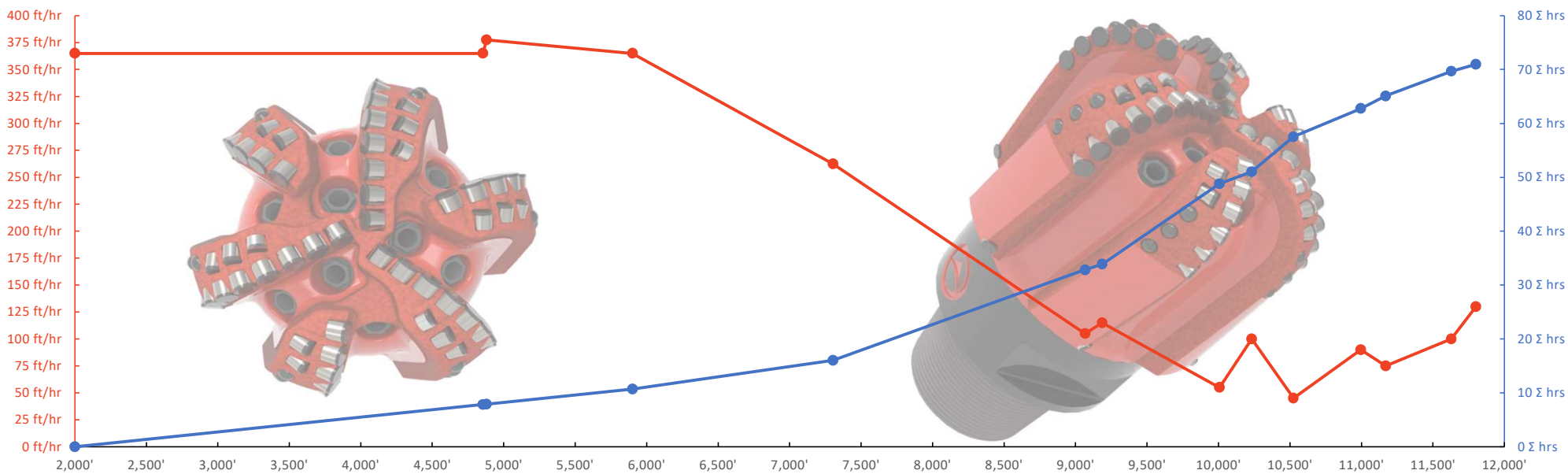
Roadmap --- Ø 9-7/8" 616-XR9(K)(BR)

Delaware --- Intermediate

Σ Measured Depth (ft)	Formation	Characteristics	WOB (KLBS)	Rotary (rpm)	Total (rpm)	Flow (GPM)	Expected ROP (ft/hr)	DOC (in/rev)	UCS kpsi	On Bottom Section (hrs)	On Bottom Cumulative (hrs)	On Bottom Cumulative ROP
						0.13						
2000 - 4855	Rustler (Salt)	Anhydrite / Limestone / Red Shale and Limestone	40 - 45	70 - 80	194 - 204	950	350 - 380	0.367	8-12	7.8	7.8	365
4855 - 4880	Lamar (Delaware)	Limey Shale	45 - 50	60 - 80	184 - 204	950	375 - 380	0.389	17-18	0.1	7.9	365
4880 - 5901	Bell Canyon	Sandstone	50 - 55	60 - 70	184 - 194	950	350 - 380	0.386	15-18	2.8	10.7	365
5901 - 7303	Cherry Canyon	Sandstone	50 - 55	60 - 65	184 - 189	950	250 - 275	0.282	18-25	5.3	16.0	331
7303 - 9067	Brushy Canyon	Shale / Sandstone / Pyritic Limestone	40 - 45	50 - 60	174 - 184	950	100 - 110	0.117	16-29	16.8	32.8	215
9067 - 9186	Upper Avalon	Massive Blocky Limestone	40 - 45	55 - 65	179 - 189	950	110 - 120	0.125	17-22	1.0	33.9	212
9186 - 10007	Avalon Lime	Massive Blocky Limestone	40 - 45	50 - 60	174 - 184	950	50 - 60	0.061	20-28	14.9	48.8	164
10007 - 10232	*1 Bone Springs Sand	Sandstone / Siltstone / Shale	45 - 50	60 - 70	184 - 194	950	80 - 120	0.106	21-25	2.3	51.0	161
10232 - 10524	2 Bone Springs Lime	Intercalated Limestone & Shale	35 - 40	55 - 60	179 - 184	950	40 - 50	0.050	24-27	6.5	57.5	148
10524 - 10996	2 Bone Springs Sand	Sandstone / Siltstone / Shale	20 - 30	40 - 50	164 - 174	950	80 - 100	0.107	24-26	5.2	62.8	143
10996 - 11170	3 Bone Springs Lime	Intercalated Limestone & Shale	30 - 35	70 - 80	194 - 204	950	70 - 80	0.075	23-26	2.3	65.1	141
11170 - 11628	Harkey Sand	Sandstone / Siltstone / Shale	45 - 50	60 - 70	184 - 194	950	90 - 110	0.106	23-25	4.6	69.7	138
11628 - 11800	3 Bone Springs Sand	Carbonate Sandstone / Siltstone / Shale	30 - 40	60 - 70	184 - 194	950	120 - 140	0.138	19-22	1.3	71.0	138
								0.150		**71.0 hrs**		**138 ROP**
										2.96 days		

* If you enter the 1 Bone Springs Sand at ≈ 100 ROP then the bit should still be green



Caprock DRILL BITS